

Neural Network Assignment 1

IMDB Movie Review Sentiment Classification

Summary Report

1. Executive Summary

This project examined how a neural network can be used to classify IMDB movie reviews as positive or negative. The analysis focused not only on prediction accuracy, but also on understanding whether a more complex neural network actually produces better results. Several model configurations were compared by changing the number of hidden layers, the number of hidden units, the loss function, the activation function, and the use of dropout.

The results showed that the simplest model tested was also the strongest. A network with one hidden layer and 16 ReLU units achieved the highest validation accuracy of **89.03%**. Deeper or larger networks produced similar results, but none improved on this performance. After the model was selected, it was trained using all 25,000 training reviews and evaluated on a separate test set of 25,000 reviews. The final test accuracy was **88.72%**, with a test loss of **0.2797**.

Overall, the analysis suggests that increasing model complexity is not always useful. For this sentiment-classification problem, a relatively simple neural network provided strong performance while remaining easier to train and interpret. The results also highlighted the importance of validation data because training performance alone can hide signs of overfitting.

2. Project Objective

The main objective of the assignment was to develop a neural network that could recognize the sentiment expressed in a movie review. Since the outcome has two categories, each review needed to be classified as either positive or negative. A second objective was to compare different neural network designs and determine which configuration performed most effectively on validation data.

Instead of assuming that additional layers or neurons would automatically improve the model, each major design change was evaluated through an experiment. This made it possible to observe how model depth, model size, activation choice, loss function, and dropout affected validation accuracy. The final architecture was selected from these comparisons rather than simply choosing the largest model.

3. Dataset and Data Preparation

The analysis used the IMDB movie review dataset. It contains **25,000 training reviews and 25,000 test reviews**, with every review labeled as positive or negative. The test set remained separate so that it could be used after model selection to evaluate performance on unseen reviews.

Because written text cannot be supplied directly to a standard dense neural network, the reviews were converted into numerical form. The analysis used the **10,000 most frequently occurring words**. Each review was represented as a binary word-presence vector, indicating whether each of these words appeared in the review. This created a consistent numerical input that could be processed by the neural network.

For the model-comparison stage, the original training set was divided into **15,000 training reviews and 10,000 validation reviews**. The training portion was used to fit each model, while the validation portion was used to compare performance. Keeping validation data separate was important because it provided a better indication of how each configuration might perform on reviews it had not directly learned from.

4. Baseline Neural Network

A baseline model was created first so that later experiments could be compared with a consistent starting point. The baseline network contained **two hidden layers with 16 units in each layer**. ReLU activation was used in the hidden layers, and a sigmoid activation was used in the output layer because the task involved a binary prediction.

The model used RMSprop as the optimizer and binary crossentropy as the loss function. Training was carried out for **20 epochs with a batch size of 512**. The best validation accuracy produced by the baseline model was approximately **88.50%**.

The training history also revealed an important issue. Training accuracy continued to improve as training progressed, while validation accuracy eventually stopped improving and validation loss began to rise. This pattern is a sign of **overfitting**. In practical terms, the model was becoming increasingly good at fitting the training examples without gaining the same benefit on data it had not seen during training.

5. Experiment 1: Number of Hidden Layers

The first major comparison examined network depth. Models with one, two, and three hidden layers were evaluated to see whether a deeper network could capture more useful patterns in the review data.

The **one-hidden-layer model achieved 89.03% validation accuracy**, which was the highest result among all experiments. The two-hidden-layer baseline achieved 88.50%, while the three-hidden-layer model reached 88.87%. Although the three-layer model performed slightly better than the baseline, it still did not exceed the simpler one-layer model.

This comparison showed that additional depth did not provide a meaningful advantage for this problem. The result is useful because it demonstrates that a more complicated architecture can add computational complexity without necessarily improving prediction quality.

6. Experiment 2: Number of Hidden Units

The next experiment focused on the width of the network by increasing the number of hidden units. Models using 32 and 64 hidden units were compared with the smaller configuration.

The model with **32 hidden units achieved 88.53%** validation accuracy, while the model with **64 hidden units achieved 88.83%**. The larger 64-unit model performed somewhat better than the 32-unit model, but both remained below the 89.03% result from the simpler one-hidden-layer configuration.

The results suggest that increasing the number of neurons gave the model additional capacity but did not translate into a meaningful improvement in validation performance. This again supports the idea that larger models are not automatically better models.

7. Experiment 3: Loss Function

Mean Squared Error (MSE) was tested as an alternative to binary crossentropy. The MSE model achieved a validation accuracy of **88.93%**, making it the second-highest result in the overall comparison.

Although the MSE result was close to the best model, it remained slightly below the 89.03% achieved by the one-hidden-layer configuration. The experiment therefore showed that changing the loss function produced only a small difference in predictive performance.

8. Experiment 4: Activation Function

The activation function was also examined by testing Tanh as an alternative to ReLU. The Tanh model achieved approximately **88.55% validation accuracy**. This was a reasonable result, but it did not improve on the stronger ReLU-based configurations.

The comparison indicates that ReLU was the more effective choice among the activation functions tested in this assignment. At the same time, the relatively small difference shows that activation choice was not the main factor driving the final result.

9. Experiment 5: Dropout

Dropout was tested because the training history showed signs of overfitting. Dropout is designed to reduce dependence on individual neurons during training, which can sometimes improve the model's ability to generalize to new data.

The 32-unit model without dropout achieved **88.53%** validation accuracy. After dropout was added, validation accuracy increased slightly to **88.68%**. This suggests that dropout provided some benefit for that particular configuration, although the improvement was small and the model still did not outperform the simpler one-hidden-layer network.

10. Overall Comparison of Models

The best validation accuracy from each experiment is shown below. The results were generally close, which makes the comparison especially useful: additional complexity often changed performance only slightly.

Model Configuration	Best Validation Accuracy
1 Hidden Layer	89.03%
MSE Loss	88.93%
3 Hidden Layers	88.87%
64 Hidden Units	88.83%
32 Units + Dropout	88.68%
Tanh Activation	88.55%
32 Hidden Units	88.53%
2 Hidden Layers - Baseline	88.50%

The strongest validation result was produced by the **one-hidden-layer model at 89.03%**. MSE loss followed closely at 88.93%, and the three-hidden-layer model reached 88.87%. The small differences across the experiments show that several configurations were capable of reasonable performance, but none of the added complexity produced a clear advantage over the simplest winning architecture.

11. Interpretation of the Results

A clear pattern emerged from the experiments: model complexity and model performance did not increase together. Adding hidden layers, increasing hidden units, changing activation functions, and applying dropout all changed the network, but the improvements were either small or nonexistent. The strongest validation result came from a relatively compact architecture.

Another important finding was the difference between training and validation behavior. A model can continue improving on the data used for training while becoming less effective on unseen examples. The observed overfitting pattern shows why validation accuracy and validation loss are important when deciding how long to train a model and which model to select.

12. Final Model Selection and Test Performance

Based on the validation comparisons, the selected final architecture used **one hidden layer with 16 ReLU units**, followed by a sigmoid output unit. RMSprop was used as the optimizer, and binary crossentropy was used as the loss function.

After the architecture was selected, the final model was trained using all **25,000 original training reviews**. It was then evaluated on the separate **25,000-review test set**. This final test provided a more realistic measure of how the selected model performed on data that had not been used for training or model selection.

The final model achieved a **test accuracy of 88.72%** and a **test loss of 0.2797**. An accuracy of 88.72% means that approximately 89 out of every 100 reviews in the test set were classified correctly. The final test result was also reasonably close to the best validation accuracy of 89.03%, showing that the selected model maintained similar performance on unseen data.

13. Key Findings

Several conclusions can be drawn from the model comparisons. First, increasing the number of hidden layers did not improve the final result. Second, increasing the number of hidden units created a larger model but did not produce a meaningful performance gain. Third, changing the loss function and activation function resulted in only modest differences. Finally, dropout slightly improved the 32-unit configuration but did not become the strongest model.

Taken together, these findings show that model selection should be based on measured validation performance rather than on model size alone. A simpler model can be preferable when it provides equal or better predictive performance because it avoids unnecessary complexity.

14. Conclusion

The project successfully developed a neural network for classifying IMDB movie reviews as positive or negative and compared several approaches for improving its performance. Among all configurations tested, the **one-hidden-layer model with 16 ReLU units achieved the highest validation accuracy of 89.03%**.

The selected architecture also performed well on the independent test set, achieving **88.72% accuracy**. This indicates that the model was able to generalize reasonably well beyond the data used during model development.

The most important lesson from the analysis is that a neural network does not need to be large or deep to perform effectively. In this case, additional layers and hidden units increased complexity without producing a meaningful improvement. Careful comparison using validation data was more valuable than simply choosing the most complicated architecture.

15. Recommendation and Possible Future Improvements

Based on the experiments, the one-hidden-layer model with 16 ReLU units is the most appropriate configuration among those tested. It provided the strongest validation result while remaining simpler than the deeper and wider alternatives.

For future development, early stopping could be used to reduce overfitting by ending training once validation performance stops improving. Additional approaches to representing the review text could also be explored. These changes may provide more useful improvements than simply increasing the number of layers or neurons.

Overall, the results demonstrate the value of balancing predictive performance with model simplicity. A well-selected, relatively simple neural network was able to achieve strong sentiment-classification performance while avoiding unnecessary architectural complexity.